

CSE 4405: DTC

Assignment: 2

Traffic Intensity, Cell Splitting & Sectorization

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Traffic Intensity, Cell Splitting and Sectorization

1. Traffic Intensity

Traffic intensity tells us how much call demand is offered to a cellular system. A cell may have good signal coverage, but if all its channels are busy, new calls will still be blocked. So traffic intensity is used to size the number of channels needed in a cell.

It is measured in Erlangs. One Erlang means one channel is busy continuously. The basic formula is:

$$A = \lambda * H$$

Here, A is traffic intensity in Erlangs, λ is the average call request rate, and H is the average call holding time. For example, if a cell gets 3000 calls per hour and each call lasts 1.76 minutes, then $\lambda = 3000/60 = 50$ calls per minute, so:

$$A = 50 * 1.76 = 88 \text{ Erlangs}$$

This does not mean exactly 88 channels are enough. Because calls arrive randomly, sometimes many users call at the same time. Therefore trunking theory, such as Erlang B, is used to decide how many channels are needed for a required Grade of Service (GoS). GoS usually means the probability that a call is blocked during the busiest hour.

2. Cell Splitting

Cell splitting is a method for increasing capacity when traffic demand becomes too high in an area. In this method, a large cell is divided into several smaller cells. Each smaller cell uses its own base station with lower antenna height and lower transmit power.

The main idea is simple: smaller cells mean more cells per unit area. Since each cell can reuse frequency channels according to the reuse plan, more cells in the same area can serve more users. This increases traffic capacity without requiring new spectrum.

The reuse ratio is:

$$Q = D / R$$

Here, D is the distance between nearest co-channel cells, and R is the cell radius. If all cells are scaled down together, both D and R decrease by the same factor, so Q remains unchanged. This means the same frequency reuse pattern can still work after splitting.

Transmit power must also be reduced in smaller cells. If the path-loss exponent is n and the cell radius is halved, the transmit power should be reduced by:

$$P_{\text{new}} = P_{\text{old}} / 2^n$$

For $n = 4$, halving the radius requires reducing power by $2^4 = 16$. The main advantages of cell splitting are higher capacity and modular growth. Its limitations are higher base-station cost, more frequent handoffs, and interference problems when small and large cells coexist.

3. Sectorization

Sectorization is another capacity improvement technique. Instead of making the cell smaller, the cell is divided into directional sectors using directional antennas. Common designs use three 120-degree sectors or six 60-degree sectors.

The purpose of sectorization is to reduce co-channel interference. With an omnidirectional antenna, a cell receives interference from all six first-tier co-channel cells. With directional antennas, the cell only receives strong interference from the co-channel cells lying in the same sector direction.

Omnidirectional antenna: about 6 dominant first-tier interferers.

120-degree sectoring: about 2 dominant first-tier interferers.

60-degree sectoring: about 1 dominant first-tier interferer.

By reducing interference, sectorization improves the signal-to-interference ratio. This allows the network to use a smaller cluster size N . Since the total number of channels S is fixed:

$$S = kN, \text{ so } k = S / N$$

If N is reduced, the number of channels per cell k increases. This is how sectorization can increase capacity. However, there is a cost: each sector gets only a fraction of the cell's channels, which reduces trunking efficiency, and more antennas are required.